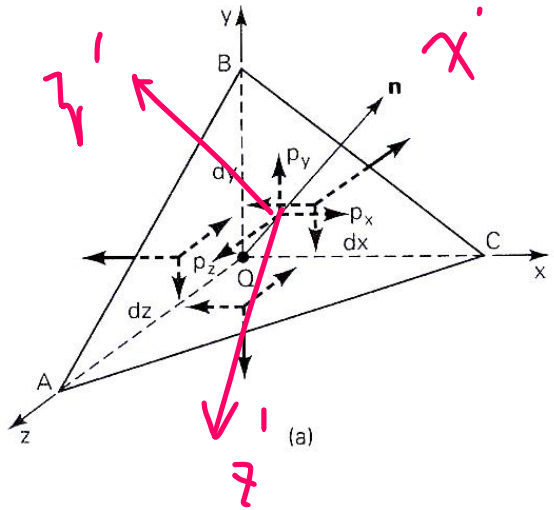
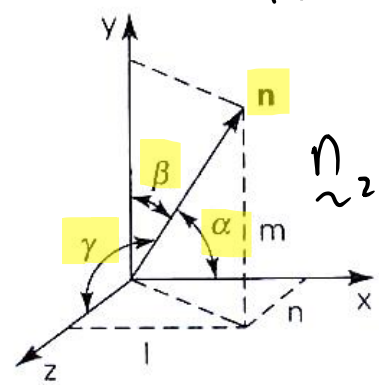
Chap. 2 Stress

2.10 Three-dimensional stress at a point

Consider a small tetrahedron isolated from a continuous medium (Fig.1.17a), subject to a general state of stress.





$\underset{\sim}{n}_1 = \cos\alpha_1 \underset{\sim}{i} + \cos\beta_1 \underset{\sim}{j} + \cos\gamma_1 \underset{\sim}{k}$

$\underset{\sim}{n}_2 = \cos\alpha_2 \underset{\sim}{i} + \cos\beta_2 \underset{\sim}{j} + \cos\gamma_2 \underset{\sim}{k}$

$\underset{\sim}{n}_3 = \cos\alpha_3 \underset{\sim}{i} + \cos\beta_3 \underset{\sim}{j} + \cos\gamma_3 \underset{\sim}{k}$

- The body forces are taken to be negligible.
- P_x, P_y & P_z are the Cartesian components of stress resultant P acting on oblique plane ABC.

① 3D problem. ①

$$\sigma_{x'} = \frac{P_{x'}}{A_{x'}} = P_x \cdot \cos\alpha_1 + P_y \cdot \cos\beta_1 + P_z \cdot \cos\gamma_1$$

$$\tau_{y'x'} = \frac{P_{y'}}{A_{x'}} = P_x \cdot \cos\alpha_2 + P_y \cos\beta_2 + P_z \cdot \cos\gamma_2$$

$$\tau_{z'x'} = \frac{P_{z'}}{A_{x'}} = P_x \cdot \cos\alpha_3 + P_y \cdot \cos\beta_3 + P_z \cos\gamma_3$$

② ②

$$P_x = \sigma_x \cdot |A_{x'}| \cdot \cos\alpha + \tau_{yx} |A_{x'}| \cdot \cos\beta + \tau_{zx} |A_{x'}| \cos\gamma$$

$$P_y = \tau_{xy} \cdot \cos\alpha + \sigma_y \cdot \cos\beta + \tau_{zy} \cdot \cos\gamma$$

$$P_z = \tau_{xz} \cdot \cos\alpha + \tau_{yz} \cdot \cos\beta + \sigma_z \cos\gamma$$

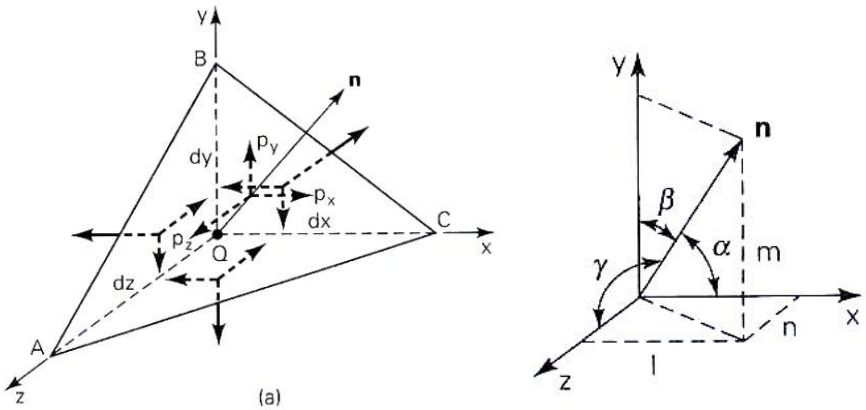
The orientation of plane ABC may be defined in terms of angles between a unit normal **n** to the plane and x, y, z directions (Fig.1.17b).

$$|\underset{\sim}{n}|\cos\alpha \rightarrow \cos\alpha = \cos(\underset{\sim}{n}, x) = \underset{\sim}{n}\cdot\underset{\sim}{i} = l$$

$$\cos\beta = \cos(\underset{\sim}{n}, y) = \underset{\sim}{n}\cdot\underset{\sim}{j} = m$$

$$\cos\gamma = \cos(\underset{\sim}{n}, z) = \underset{\sim}{n}\cdot\underset{\sim}{k} = n$$

$$\underset{\sim}{n} = l\,\underset{\sim}{i} + m\,\underset{\sim}{j} + n\,\underset{\sim}{k}$$



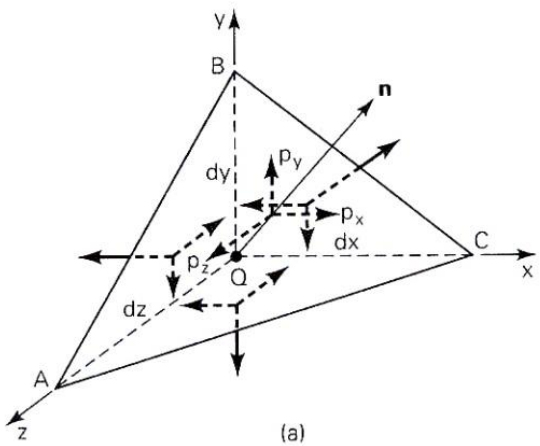
Here, i, j, & k are unit vectors in the x, y, & z directions, respectively.

These direction cosines are related by

$$l^2 + m^2 + n^2 = 1 \quad \leftarrow |\underset{\sim}{n}| = 1$$

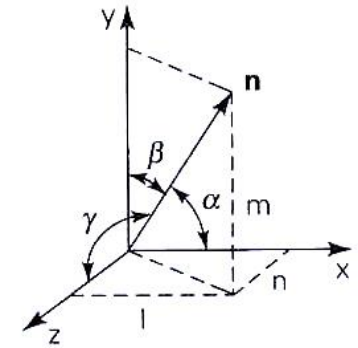
The area of the perpendicular plane QAB, QAC, QBC may now be expressed in terms of A, the area of ABC, and the direction cosines:

$$\boxed{A_{QAB} = A_x = \underset{\sim}{A} \cdot \underset{\sim}{i} = A\underset{\sim}{n} \cdot \underset{\sim}{i} = A(l\,\underset{\sim}{i} + m\,\underset{\sim}{j} + n\,\underset{\sim}{k}) = Al}$$



The other two areas

$$\boxed{A_{QAC} = Am, \;\; A_{QBC} = An}$$



From the equilibrium of x,y,z-direction force

$$P_x = \sigma_x l + \tau_{yx} m + \tau_{zx} n$$

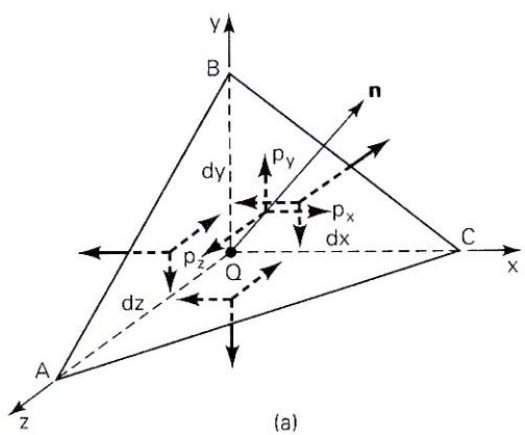
$$P_y = \tau_{xy} l + \sigma_y m + \tau_{zy} n \qquad (*)$$

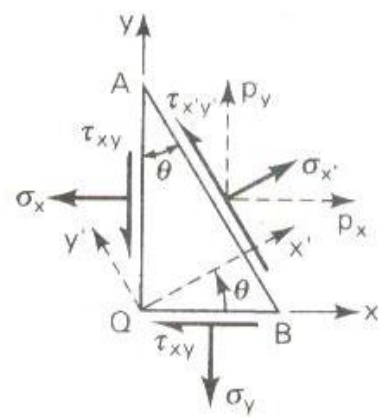
$$P_z = \tau_{xz} l + \tau_{yz} m + \sigma_z n$$

<=

$$\boxed{\begin{aligned} P_x &= \sigma_x A_x + \tau_{yx} A_y + \tau_{zx} A_z \\ &= \sigma_x A l + \tau_{yx} A m + \tau_{zx} A n \\ &= \sigma_x l + \tau_{yx} m + \tau_{zx} n \end{aligned}}$$

- The stress resultant on A is thus determined on the basis of known stresses.





$$\boxed{\begin{aligned} p_x &= \sigma_x \cos\theta + \tau_{xy} \sin\theta \\ p_y &= \tau_{xy} \cos\theta + \sigma_y \sin\theta \end{aligned}}$$

Consider a Cartesian coordinate system x', y', z', wherein x' coincides with **n** and y', z' lie on an oblique plane.

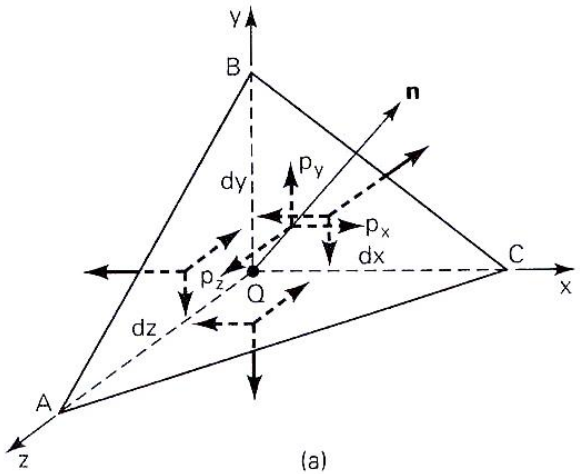
- The x'y'z' & xyz systems are related by $l_1 = \cos(x', x)$, $m_1 = \cos(x', y)$, $n_1 = \cos(x', z)$ and so on.

For the notation see Table1.2.

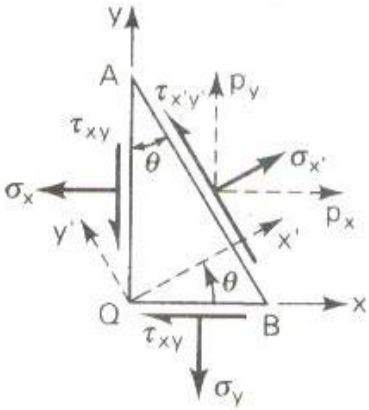
	x	**y**	**z**
X'	l_1	m_1	n_1
Y'	l_2	m_2	n_2
Z'	l_3	m_3	n_3

The normal stress $\sigma_{x'}$ is found by projecting P_x, P_y, & P_z in the x' direction and adding :

$$\sigma_{x'} = p_x l_1 + p_y m_1 + p_z n_1$$
$$\tau_{y'x'} = p_x l_2 + p_y m_2 + p_z n_2 \qquad (**)$$
$$\tau_{z'x'} = p_x l_3 + p_y m_3 + p_z n_3$$



$$\underset{\sim}{p} \cdot \underset{\sim}{n_x} = p_x l_1 + p_y m_1 + p_z n_1$$
$$\underset{\sim}{p} \cdot \underset{\sim}{n_y} = p_x l_2 + p_y m_2 + p_z n_2$$



$$\sigma_{x'} = p_x \cos\theta + p_y \sin\theta$$
$$\tau_{x'y'} = p_y \cos\theta - p_x \sin\theta$$

Eqn (*) & (**) are combined to yield

$$\sigma_{x'} = \sigma_x l_1^2 + \sigma_y m_1^2 + \sigma_z n_1^2 + 2(\tau_{xy} l_1 m_1 + \tau_{yz} m_1 n_1 + \tau_{zx} l_1 n_1)$$

$$\tau_{x'y'} = \sigma_x l_1 l_2 + \sigma_y m_1 m_2 + \sigma_z n_1 n_2 + \tau_{xy}(l_1 m_2 + m_1 l_2) + \\ + \tau_{yz}(m_1 n_2 + n_1 m_2) + \tau_{xz}(n_1 l_2 + l_1 n_2)$$

$$\tau_{x'z'} = \sigma_x l_1 l_3 + \sigma_y m_1 m_3 + \sigma_z n_1 n_3 + \tau_{xy}(l_1 m_3 + m_1 l_3) + \\ + \tau_{yz}(m_1 n_3 + n_1 m_3) + \tau_{xz}(n_1 l_3 + l_1 n_3)$$

$$\sigma_{x'} = p_x l_1 + p_y m_1 + p_z n_1$$
$$\tau_{y'x'} = p_x l_2 + p_y m_2 + p_z n_2$$
$$\tau_{z'x'} = p_x l_3 + p_y m_3 + p_z n_3$$

$$P_x = \sigma_x l + \tau_{yx} m + \tau_{zx} n$$
$$P_y = \tau_{xy} l + \sigma_y m + \tau_{zy} n$$
$$P_z = \tau_{xz} l + \tau_{yz} m + \sigma_z n$$

▼ The remaining components are found by considering those planes perpendicular to the oblique plane.

See Eqns.(1.24d),(1.24e),(1.24f).

$\sigma_{y'}, \sigma_{z'}, \tau_{y'z'}$

- It is interesting to note that, because x', y', z' are orthogonal, the nine direction cosines must satisfy trigonometric relations of the following form:

$$l_i^2 + m_i^2 + n_i^2 = 1, \quad i = 1,2,3 \qquad \leftarrow |n_i| = 1, \quad i = 1,2,3$$

$$\left.\begin{aligned} l_1 l_2 + m_1 m_2 + n_1 n_2 &= 0 \\ l_2 l_3 + m_2 m_3 + n_2 n_3 &= 0 \\ l_1 l_3 + m_1 m_3 + n_1 n_3 &= 0 \end{aligned}\right\}$$

Orthogonality of each normal vector

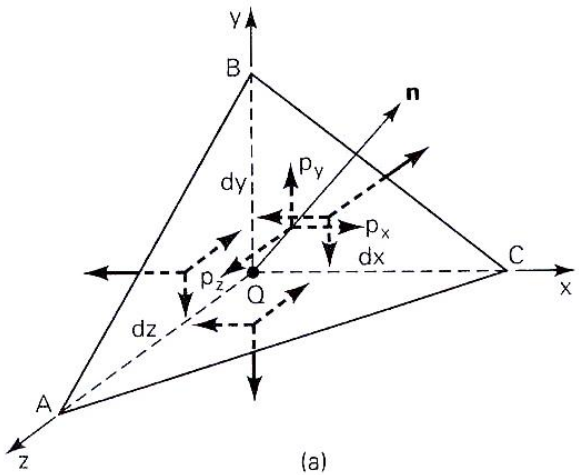
$$\underset{\sim}{n}_i \cdot \underset{\sim}{n}_j = 0 \quad if \quad i \neq j \quad where \quad i = 1,2,3, \quad j = 1,2,3$$

2.11 Principal stresses in three dimensions

Consider an oblique x' plane.
The normal stress acting on this plane is given by

$$\sigma_{x'} = \sigma_x l^2 + \sigma_y m^2 + \sigma_z n^2 + 2(\tau_{xy} lm + \tau_{yz} mn + \tau_{zx} ln)$$

To determine extreme or stationary values of $\sigma_{x'}$, we examine the variation of $\sigma_{x'}$ relative to the direction cosines.



$$\boxed{\sigma_{x'} = \sigma_x l^2 + \sigma_y m^2 + \sigma_z n^2 + 2(\tau_{xy} lm + \tau_{yz} mn + \tau_{zx} ln)}$$

- From $l^2+m^2+n^2=1$, only l & m may be regarded as independent variables

$$\boxed{\begin{aligned} \sigma_{x'} &= p_x l_1 + p_y m_1 + p_z n_1 \\ \tau_{y'x'} &= p_x l_2 + p_y m_2 + p_z n_2 \\ \tau_{z'x'} &= p_x l_3 + p_y m_3 + p_z n_3 \end{aligned}}$$

$$\frac{\partial \sigma_{x'}}{\partial l} = 0, \quad \frac{\partial \sigma_{x'}}{\partial m} = 0$$

Using eqn. (**) of Ch. 1.12

$$p_x + p_z \frac{\partial n}{\partial l} = 0, \quad p_y + p_z \frac{\partial n}{\partial m} = 0 \qquad (*^3)$$

- From $n^2=1-l^2-m^2$, we have $\dfrac{\partial n}{\partial l} = -\dfrac{l}{n}$ & $\dfrac{\partial n}{\partial m} = -\dfrac{m}{n}$

Introducing these into Eqn.(*3), the follow relationships between the components of **p** and **n** are determined:

$$\frac{p_x}{l} = \frac{p_y}{m} = \frac{p_z}{n}$$

$$\frac{\partial n}{\partial l} = -\frac{l}{n} \quad \& \quad \frac{\partial n}{\partial m} = -\frac{m}{n}$$

$$p_x + p_z \frac{\partial n}{\partial l} = 0, \quad p_y + p_z \frac{\partial n}{\partial m} = 0$$

$$\Rightarrow \sigma_x + \tau_{yx}\frac{m}{l} + \tau_{zx}\frac{n}{l} = \tau_{xy}\frac{l}{m} + \sigma_y + \tau_{zy}\frac{n}{m}$$
$$= \tau_{xz}\frac{l}{n} + \tau_{yz}\frac{m}{n} + \sigma_z \quad \Rightarrow \sigma_x = \sigma_y = \sigma_z \Rightarrow \sigma_p$$

– On one oblique surface, one σ_p

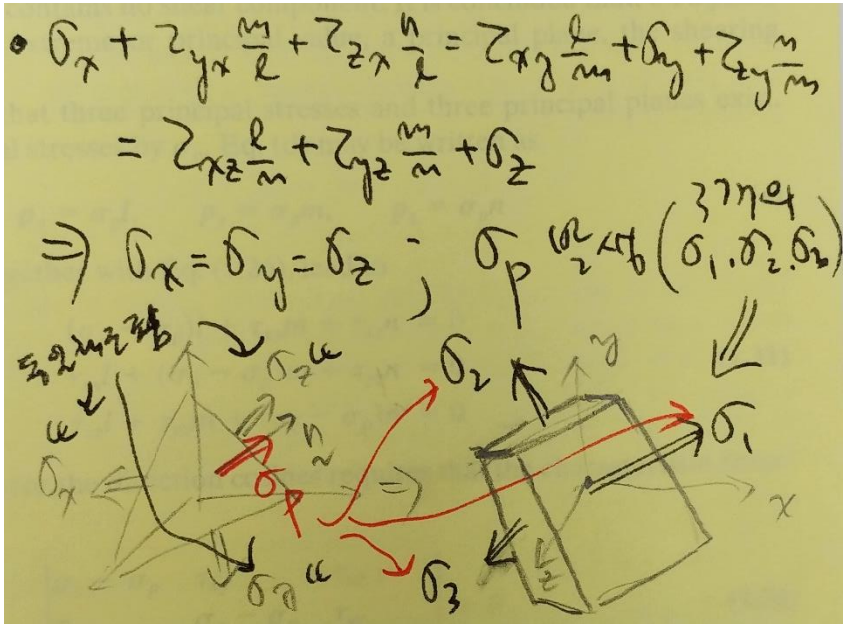
$$P_x = \sigma_x l + \tau_{yx} m + \tau_{zx} n$$
$$P_y = \tau_{xy} l + \sigma_y m + \tau_{zy} n$$
$$P_z = \tau_{xz} l + \tau_{yz} m + \sigma_z n$$

▼ These proportionalities indicate that

① The stress resultant must be parallel to the unit normal (ex, P_x=C*l, P_y=C*m, P_z=C*n => P=C*n; parallel to the unit normal)

② contains no shear component

③ On an oblique plane for which $\sigma_{x'}$ has an extreme or principal value, the shear stress vanishes on the x, y & z plane



$$\sigma_x = \sigma_y = \sigma_z \Rightarrow \sigma_p$$

The maximum stress on the oblique surface should be σ_p

Denoting the principal stresses by σ_p.

(If only principal stresses exist,)

$$P_x = \sigma_p l, \qquad P_y = \sigma_p m, \qquad P_z = \sigma_p n$$

<=>

$$\boxed{\begin{aligned} P_x &= \sigma_x l + \tau_{yx} m + \tau_{zx} n \\ P_y &= \tau_{xy} l + \sigma_y m + \tau_{zy} n \\ P_z &= \tau_{xz} l + \tau_{yz} m + \sigma_z n \end{aligned}}$$

Then inversely these expressions, together with Eqn.(*) of Ch.1.12 (Eq.(1.22)), lead to

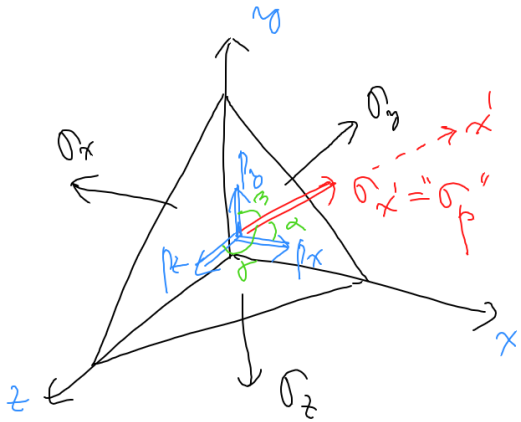
$$(\sigma_x - \sigma_p) l + \tau_{yx} m + \tau_{zx} n = 0$$

$$\tau_{xy} l + (\sigma_y - \sigma_p) m + \tau_{zy} n = 0$$

$$\tau_{xz} l + \tau_{yz} m + (\sigma_z - \sigma_p) n = 0$$

$$\Updownarrow$$

$$\boxed{\begin{aligned} P_x &= \sigma_x l + \tau_{yx} m + \tau_{zx} n \\ P_y &= \tau_{xy} l + \sigma_y m + \tau_{zy} n \\ P_z &= \tau_{xz} l + \tau_{yz} m + \sigma_z n \end{aligned}}$$



$P_x = \sigma_p \cdot A_{x'} \cos\alpha$ ← l

$P_y = \sigma_p A_{x'} m$

$P_z = \sigma_p A_{x'} \cdot n$

A nontrivial solution

$$\begin{vmatrix} \sigma_x - \sigma_p & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y - \sigma_p & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z - \sigma_p \end{vmatrix} = 0 \qquad (*) \Leftarrow \begin{bmatrix} \sigma_x - \sigma_p & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y - \sigma_p & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z - \sigma_p \end{bmatrix} \begin{bmatrix} l \\ m \\ n \end{bmatrix} = 0$$

Expanding Eq (*) lead, to

$$\sigma_p^3 - I_1 \sigma_P^2 + I_2 \sigma_P - I_3 = 0 \qquad (**)$$

"σ_1 & σ_2 & σ_3"

where

$$I_1 = \sigma_x + \sigma_y + \sigma_z$$

$$I_2 = \sigma_x \sigma_y + \sigma_x \sigma_z + \sigma_y \sigma_z - \tau_{xy}^2 - \tau_{yz}^2 - \tau_{xz}^2$$

$$I_3 = \begin{vmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{vmatrix}$$

"Invariants"

independant from coordinate change!!

- The three roots of the stress cubic equation(**) are the principal stresses, corresponding to which are three sets of direction cosines.

$$\sigma_p^3 - I_1\sigma_P^2 + I_2\sigma_P - I_3 = 0$$

- The principal stresses are the real eigenvalues of the stress tensor.

▼ It is clear that the principal stresses are independent of the orientation of the original coordinate system.

It follows that the coefficients I_1, I_2, & I_3 must likewise be independent of x, y, & z.

ex) $I_1 = \sigma_{x'} + \sigma_{y'} + \sigma_{z'} = \sigma_x + \sigma_y + \sigma_z$

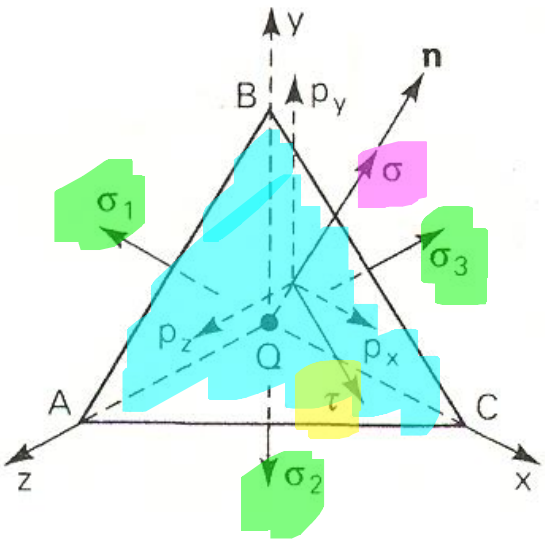
for plane σ

$$I_1 = \sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y$$

$$I_2 = I_3 = \sigma_{x'}\sigma_{y'} - \tau_{x'y'}^2 = \sigma_x\sigma_y - \tau_{xy}^2$$

2.12 Normal & shear stresses on an oblique plane

Given the principal stresses acting on perpendicular plane
-> Determine the shearing and normal stresses acting on an arbitrary oblique plane of a tetrahedron.



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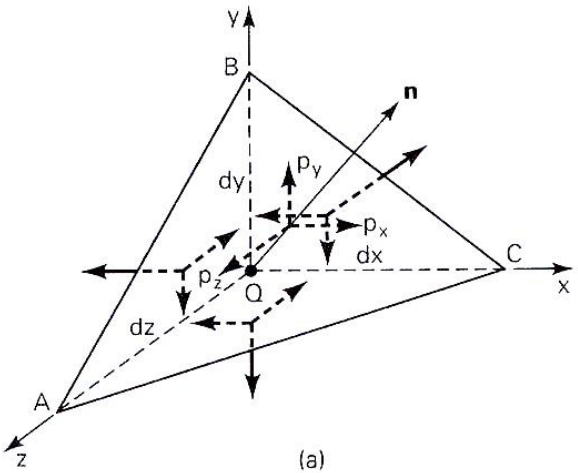


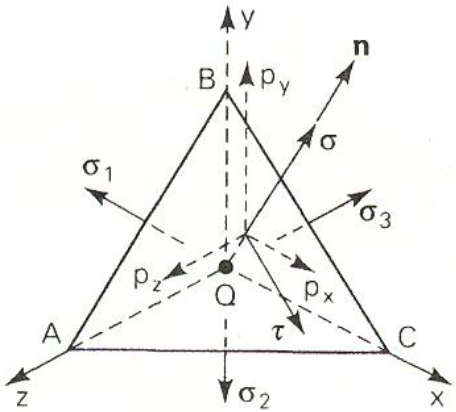
$\sigma_1 \neq \sigma_2 \neq \sigma_3$ => No principal stress on the surface

Denoting the direction cosines of plane ABC by l, m, and n, Eqn.(*) of ch.12 (Eqn.(1.22)) with $\sigma_x = \sigma_1$, $\tau_{xy} = \tau_{xz} = 0$, reduce to

$$P_x = \sigma_1 l,\ P_y = \sigma_2 m,\ P_z = \sigma_3 n \quad \Leftrightarrow \quad \begin{aligned} P_x &= \sigma_x l + \tau_{yx} m + \tau_{zx} n \\ P_y &= \tau_{xy} l + \sigma_y m + \tau_{zy} n \\ P_z &= \tau_{xy} l + \tau_{yx} m + \sigma_z n \end{aligned}$$

Stress resultant P is related to the principal stresses and the stress components on the oblique plane by the expression.

$$P^2 = \sigma_1^2 l^2 + \sigma_2^2 m^2 + \sigma_3^2 n^2 = \sigma^2 + \tau^2 \qquad \text{(a)}$$



The normal stress σ on this plane is found as

$$\sigma = \sigma_1 l^2 + \sigma_2 m^2 + \sigma_3 n^2$$

<= $$\boxed{\sigma_{x'} = \sigma_x l^2 + \sigma_y m^2 + \sigma_z n^2 + 2(\tau_{xy} lm + \tau_{yz} mn + \tau_{zx} ln)}$$

Substitution of this expression into Eq.(a) lead to

$$\tau^2 = \sigma_1^2 l^2 + \sigma_2^2 m^2 + \sigma_3^2 n^2 - (\sigma_1 l^2 + \sigma_2 m^2 + \sigma_3 n^2)^2$$

$$\boxed{P^2 = \sigma_1^2 l^2 + \sigma_2^2 m^2 + \sigma_3^2 n^2 = \sigma^2 + \tau^2}$$

$$\tau^2 = \sigma_1^2 l^2 + \sigma_2^2 m^2 + \sigma_3^2 n^2 - (\sigma_1 l^2 + \sigma_2 m^2 + \sigma_3 n^2)^2$$

Expanding and using the expression $1-l^2=m^2+n^2$, $1-n^2=l^2+m^2$, the following result is obtained for the shearing stress τ on the oblique plane:

$$\tau = \left[(\sigma_1-\sigma_2)^2 l^2 m^2 + (\sigma_2-\sigma_3)^2 m^2 n^2 + (\sigma_3-\sigma_1)^2 n^2 l^2\right]^{\frac{1}{2}}$$

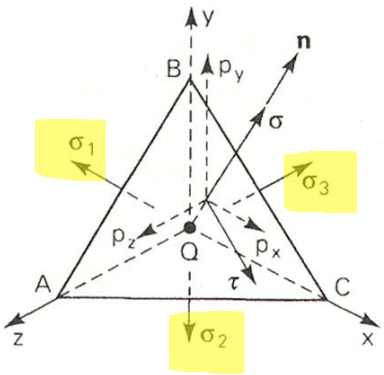
▼ If the principal stresses are all equal, the shear stress vanishes (Hydrostatic stress state) regardless of the choices of the direction cosines.

$$\tau = 0$$

$$\sigma = \sigma_1 (l^2 + m^2 + n^2) = \sigma_1$$

▼ The stresses acting on octahedral plane

$$l = m = n = \frac{1}{\sqrt{3}}$$



Octahedral normal stress

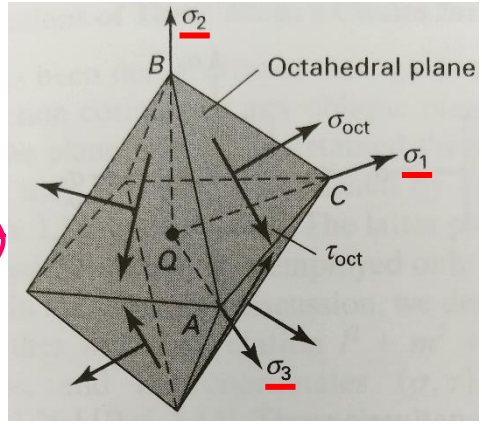
$$\sigma_{oct} = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3) \quad <= \quad \boxed{\sigma = \sigma_1 l^2 + \sigma_2 m^2 + \sigma_3 n^2}$$

Octahedral shear stress

$$\tau_{oct} = \frac{1}{3}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]^{1/2}$$

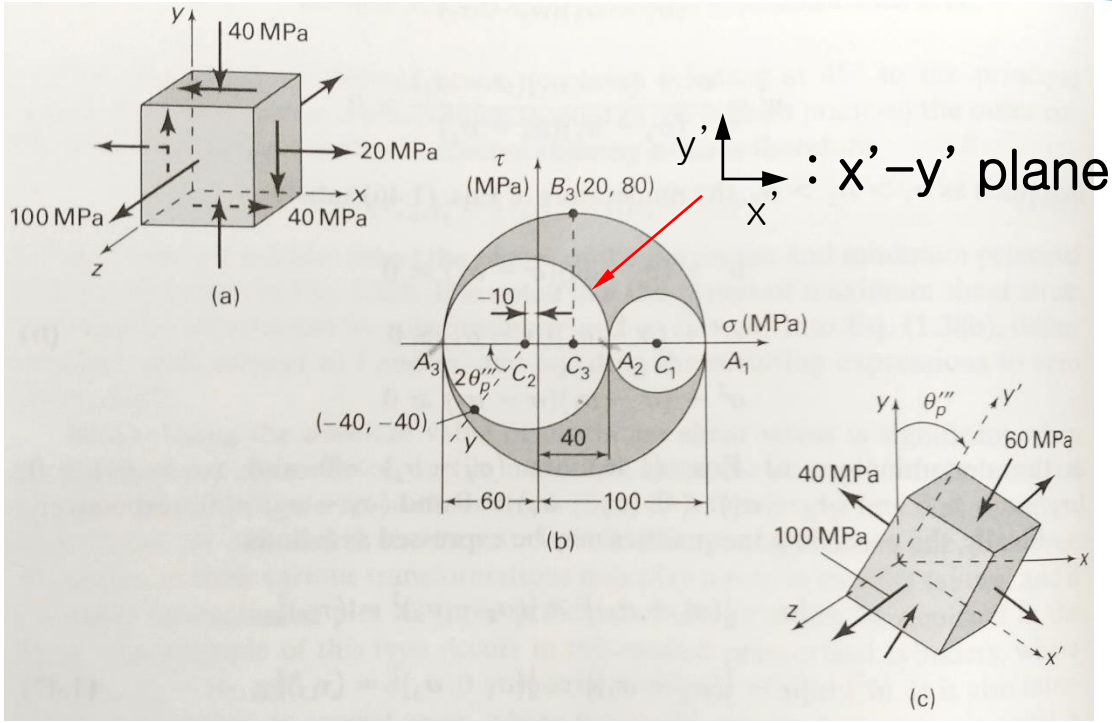
$$\Uparrow$$

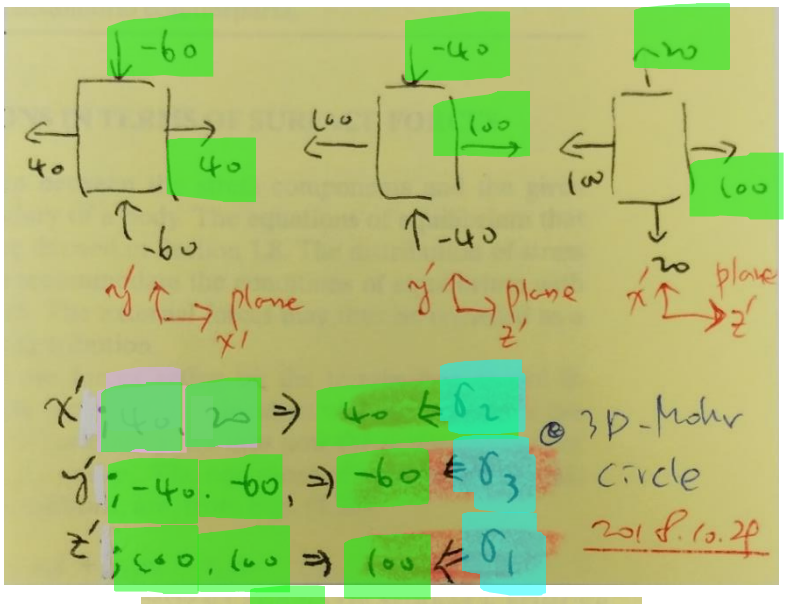
$$\boxed{\tau = \left[(\sigma_1 - \sigma_2)^2 l^2 m^2 + (\sigma_2 - \sigma_3)^2 m^2 n^2 + (\sigma_3 - \sigma_1)^2 n^2 l^2\right]^{\frac{1}{2}}}$$

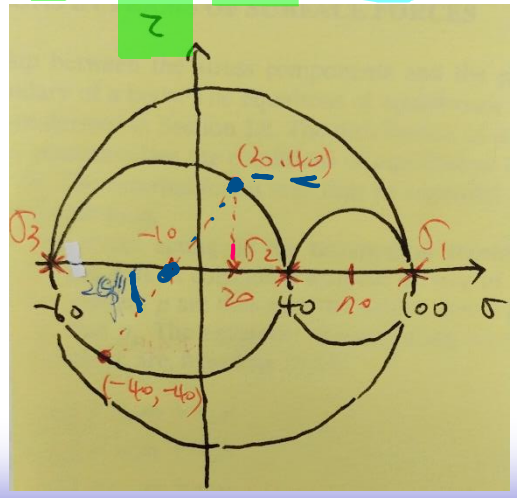


octahedron

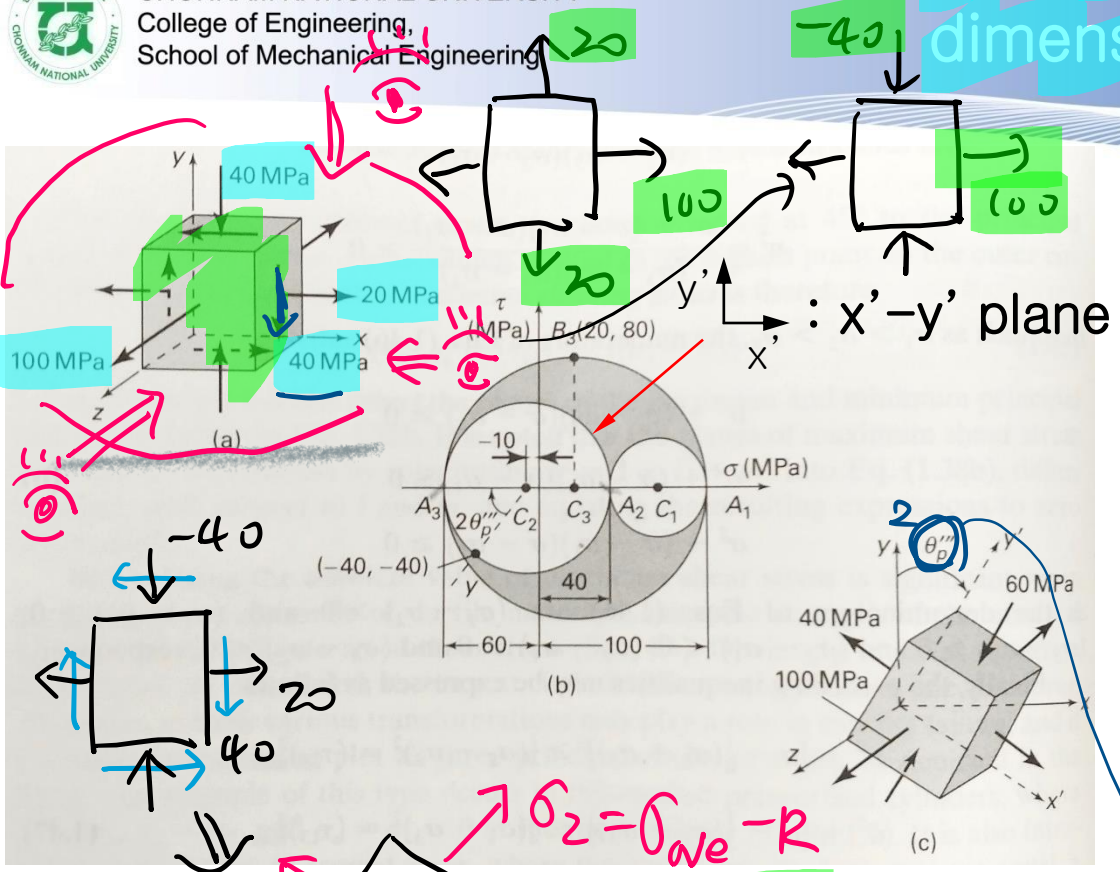
2.13 Mohr's circle for three dimensional stress



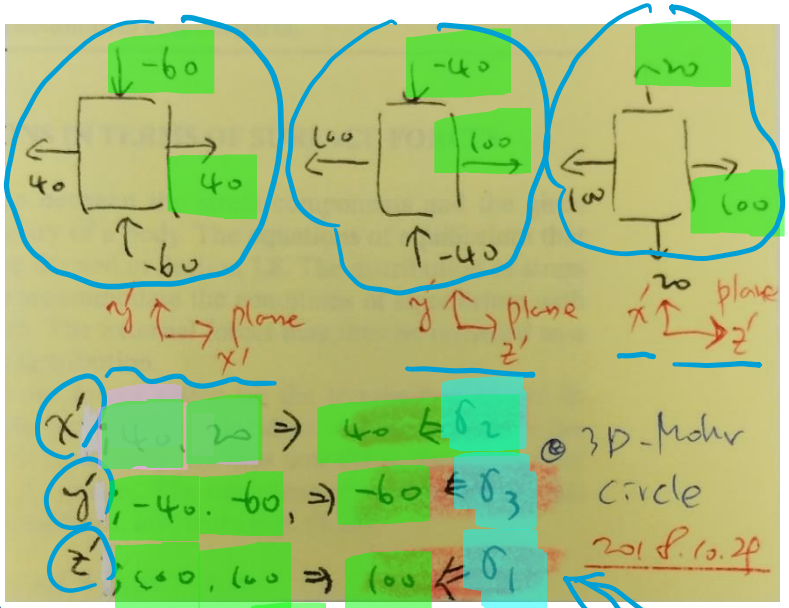


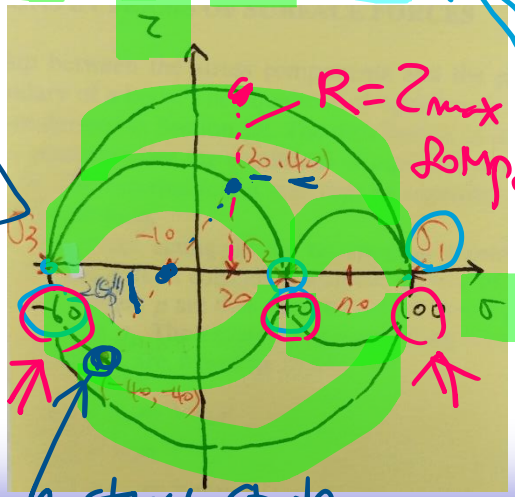


2.13 Mohr's circle for three dimensional stress



: x'−y' plane





$\sigma_2 = \sigma_{ave} - R = -60$

$\sigma_1 = \sigma_{ave} + R = 40$

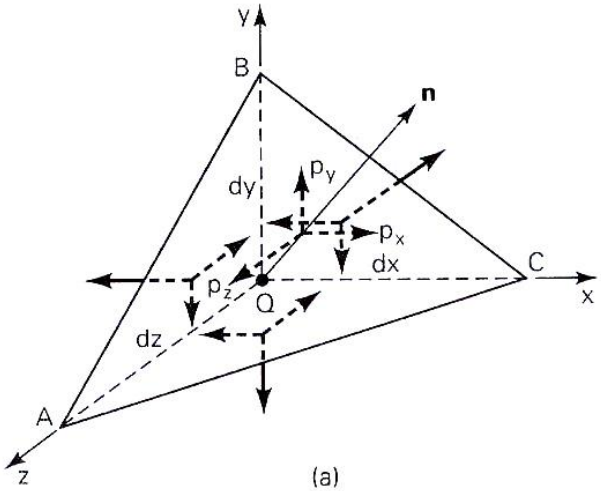
$\frac{\sigma_x + \sigma_y}{2}$ $\sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

▼ READ Example 1.8

2.14 Boundary conditions in terms of surface forces

We now consider the relationship between the stress components and the given surface forces acting on the boundary of a body.

▼ The external forces may be regarded as a continuation of the internal stress distribution.

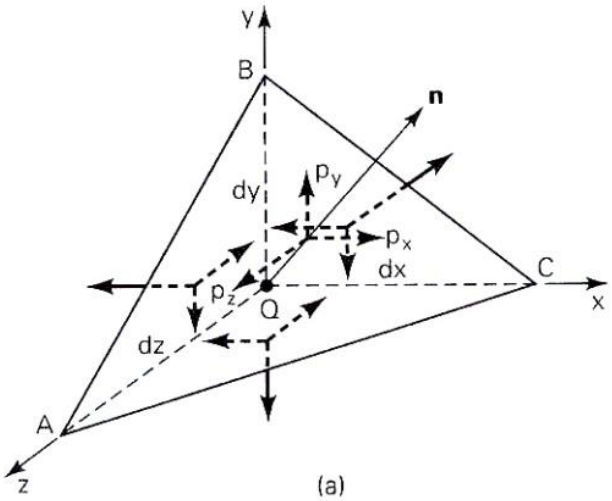


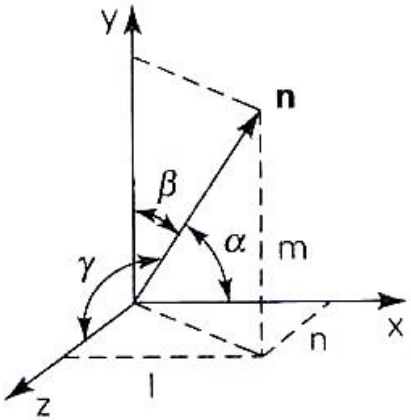
Consider the equilibrium of the forces acting on the tetrahedron shown in Fig.1.17a & assume that oblique face ABC is coincident with the surface of the body.

Now the stress resultant **p** are the surface forces per unit area, or Surface tractions.

The equations of equilibrium for this element, representing boundary conditions are,

$$p_x = \sigma_x l + \tau_{xy} m + \tau_{xz} n$$
$$p_y = \tau_{xy} l + \sigma_y m + \tau_{yz} n$$
$$p_z = \tau_{xz} l + \tau_{yz} m + \sigma_z n$$



$$p_x = \sigma_x l + \tau_{xy} m + \tau_{xz} n$$
$$p_y = \tau_{xy} l + \sigma_y m + \tau_{yz} n$$
$$p_z = \tau_{xz} l + \tau_{yz} m + \sigma_z n$$

- If the boundary is a plane with an x-directed surface normal. Eqs. above give $p_x = \sigma_x$, $p_y = \tau_{xy}$, $p_z = \tau_{xz}$; under these circumstances, the applied surface force components p_x, p_y, & p_z are balanced by σ_x, τ_{xy}, & τ_{xz}, respectively. (l=1, m=0, n=0 <= α=0, β=90, γ=90)